MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3310 ROUGHER AND SCAVENGER FLOTATION
DESIGN OF CRANE RAIL FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to Prepare for design of **Crane Rail Foundation (3310)**. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Load Data Provided.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Gantry Crane Drawing (Doc. No. MD-216-3300-EG-ME-GAD-V557-1003) attached in Attachment 1. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5 CONCRETE FOUNDATIONS

5.1 Design

Design of foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar: ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

5.3 Foundations for Aboveground Tanks

Foundation shall be Earth Foundation with a Concrete Ring wall. The design calculations are prepared as per recommendation of standard API 650 - Annex B.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall	50
	Foundation	50

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5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m^2 at 1.5m depth below EGL (ie. RL 911.50m) refer Table 5 - BH-10 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

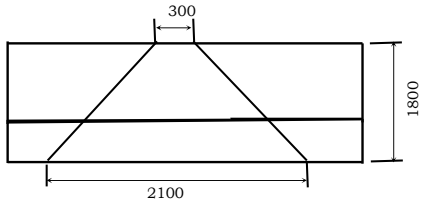
6. DESIGN CALCULATION FOR FOUNDATION

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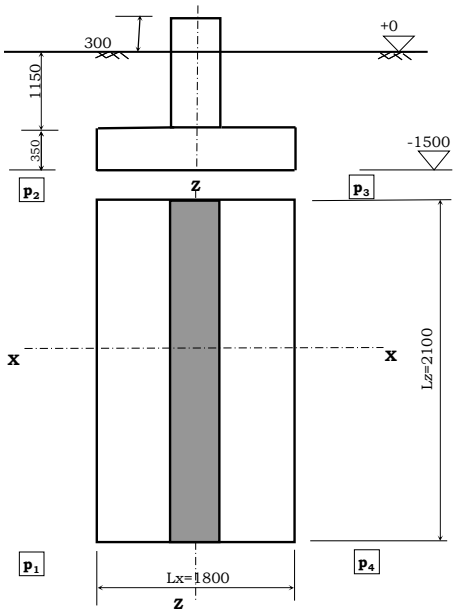
DESIGN OF FOUNDATION FOR CRANE

	Along-X	Along-Z
Size of Wall		0.45m
Size of fdn. raft:	1.80m	2.10m
Thickness of raft:	0.35 m	
Original Gr. Level:	0.00 m	
Finished Gr. Level:	0.00 m	
U/S of B/Pl Above GL:	0.30 m	
U/S level of fdn:	-1.50 m	
Unit weight of conc:	25 kN/m^3	
Unit weight of soil:	18 kN/m^3	



Calculation of Weight of Foundation :

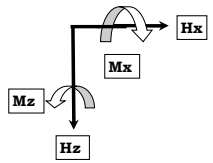
Weight of raft =	(1.8*2.1*0.35*25) =	33.08KN
Weight of Wall =	(1.45*2.1*0.35*25) =	26.64KN
Weight of soil =	((1.8*2.1)-(2.1*0.45))*1.15*18 =	58.68KN
	Total =	118.40KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%
Dia. of bars provided wall vertical =	16mm

Effective depth of slab along X direction = 288mm = 0.288m
Effective depth of slab along Z direction = 294mm = 0.294m
Effective depth of Wall = 392mm = 0.392m
Clear projection along X direction = 0.675m



H_x=Force along X-X direction

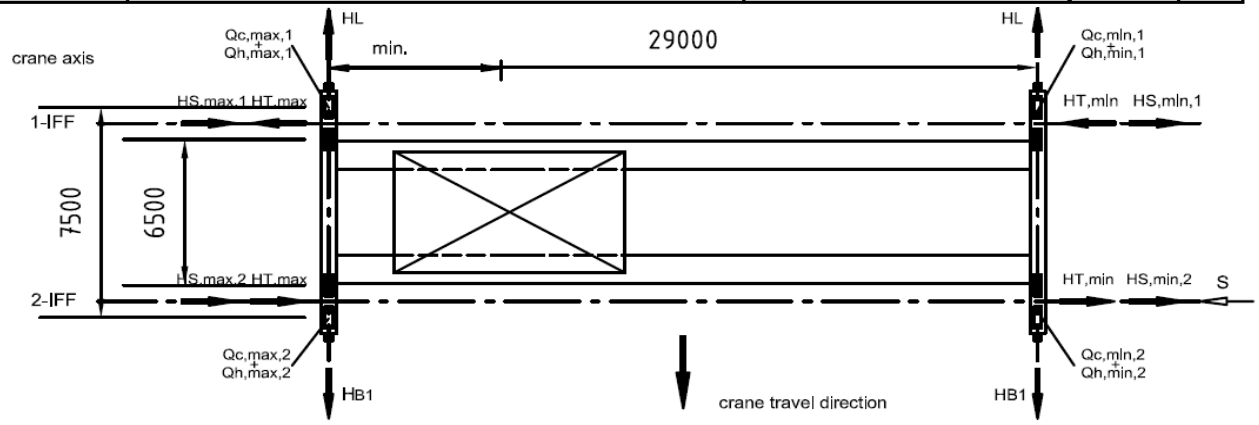
Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

Basic Load Cases :

Wheel loads (vertical)	Force component from the mass of the crane and the hoist(s) per crane axis	Qc,min,1	30.7	Qc,max,1	41.6	[kN]
		Qc,min,2	37.9	Qc,max,2	48.8	[kN]
	Force component from mass of hoist load per crane axis	Qh,min,1	5.9	Qh,max,1	29.7	[kN]
		Qh,min,2	5.8	Qh,max,2	29.7	[kN]
Lateral loads (horizontal)	Force from the acceleration of the crane with hoist load (mass force)	HT,min	5.7	HT,max	7.8	[kN]
	skewing force (coefficient of frictional contact <= 0,3)	S			53.2	[kN]
	Horizontal force from skewing per crane axis	HS,min,1	0.0	HS,max,1	0.0	[kN]
		HS,min,2	3.3	HS,max,2	49.9	[kN]
Longitudinal loads (horizontal) (per runway beam)	Force from the acceleration of the crane with hoist load (mass force)	HL			11.8	[kN]
	Force from impact on buffer (collision force) (motion limiter taken into account)	HB1			31.1	[kN]



Loads on Rail :-

Maximum Vertical Load (Q_C max 2 + Q_H max 2)= 48.8 + 29.7 kN

= 78.5 kN

Maximum Horizontal Load(H_S max 2 + H_T max 2)= 49.9 + 7.8 kN

57.7 kN

Sl. No	Basic Load Case	V (kN)	Hx (kN)
1	Dead Load (D)	0.0	0.0
2	Crane Load (Cr)	78.5	57.7

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D + Cr	78.5	57.7	0.0	0.0	0.0	196.90	57.70	103.86	0.00	0.00
Ultimate Conditions											
301	1.2 D + 1.6 Cr	125.6	92.3	0.0	0.0	0.0	244.00	92.32	166.18	0.00	0.00

Check for overturning and sliding:

Load Cases:		Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + Cr	196.90	57.70	103.86	0.00	0.00	1.71		1.54	

DESIGN OF FOUNDATION FOR CRANE 0

Load Combination & Calculation of Base pressure

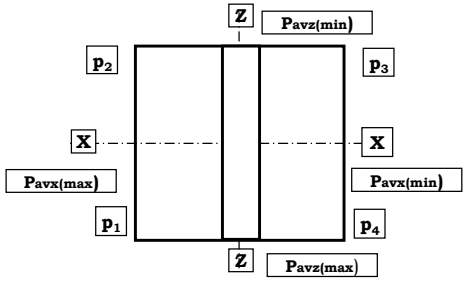
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m²

Net safe bearing capacity of soil for DL, LL= 375 kN/m² For Wind load/Seis. load cas 468.75 kN/m²

Gross safe bearing capacity of soil for DL, LL= 402 kN/m² For Wind load/Seis. load cas 495.75 kN/m²



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m²)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2.Lz$	6.Mx/ $Lx.(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	D + Cr	196.90	103.86	0.00	52.09	91.59	0.00	183.17	183.17	0.00	0.00	156.17	183.17	0.00	91.59	91.59	**O.K**
301	1.2 D + 1.6 Cr	244.00	166.18	0.00	64.55	146.54	0.00	266.75	266.75	0.00	0.00	239.75	266.75	0.00	133.38	133.38	

DESIGN OF FOUNDATION FOR CRANE

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

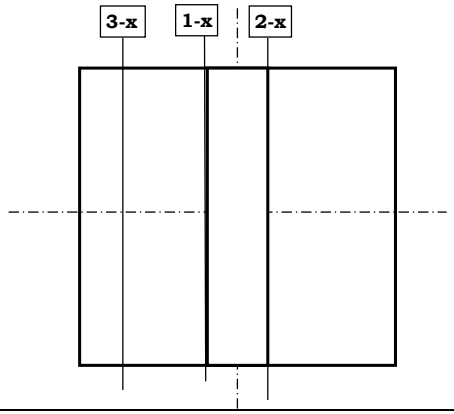
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((3.15+2*0.5*0.294)+(0.45+2*0.5*0.294))*2

= 8.376 m

Punching Area= ((3.15+2*0.5*0.294)*(0.45+2*0.5*0.294))

= 2.56 m^2 Punching shear stress to be checked



Downward pressure due to self weight of raft and soil = (0.35*25+1.15*0.35*18)

= 29.45 kN/m^2

Area of footing, A_f= 3.78 m^2

DESIGN OF FOUNDATION FOR CRANE

Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
301	1.2 D + 1.6 Cr	166.72	100.03	209.40	46.46	0.89	80.74	-316.90

Overhang = 0.675 m

Effective depth = 0.288 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	46.46	0.56	0.75	0.002	518.76	518.40	12ϕT	218.01	300	200	565.49
At top	24.00	0.29	0.39	0.001	0.00	259.20	12ϕT	436.33	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi Vc)/bd$ = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 80.74 kN/m

Shear stress , τ_v= (80.74*1000)/(1000*288)

= 0.28 N/mm^2 **OK**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= -80.18 kN

Strength of Concrete in shear Vc1 = 3058.13 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 3196.39 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 4757.10 kN **Table 22.6.5.2_ (c)**

Design Vc= 3058.13 kN

Allowable shear stress, $(\phi Vc)/bd$ = 0.95 N/mm^2

Design punching shear stress, τ = (-80.18*1000)/(8376*294)

= -0.03 N/mm^2 **No shear reinforcement reqd.**

DESIGN OF FOUNDATION FOR CRANE

Calculation of distribution reinf in Z direction :

Effective depth = 0.294 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	12.00	0.14	0.19	0.000	129.98	264.60	12ϕT	427.43	300	250	452.39
At top	12.00	0.14	0.19	0.000	129.98	264.60	12ϕT	427.43	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 367.78 kN/m

Shear stress , τ_v= (367.78*1000)/(1000*294)

= 1.25 N/mm^2 **OK**

Calculation of reinf in Wall :

Effective depth = 0.392 m

Load Case	Load Combination	$B.M_{wall}$ (kN-m)	$S.F_{wall}$ kN/m
301	1.2 D + 1.6 Cr	63.74	43.96

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At Vertical	63.74	0.41	0.55	0.001	0.00	576.00	16ϕT	349.07	300	200	1005.31
At Horizontal	50.00	0.33	0.43	0.001	0.00	345.60	12ϕT	327.25	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 43.96 kN/m

Shear stress , τ_v= (43.96*1000)/(1000*288)

= 0.15 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
- At Bottom **12ϕT** **200** c/c
- At Top **12ϕT** **250** c/c

Along Z Direction :
- At Bottom **12ϕT** **250** c/c
- At Top **12ϕT** **250** c/c

SUMMARY OF REINFORCEMENT IN WALL:

Vertical Direction:
16ϕT **200** c/c

Horizontal Direction :
12ϕT **200** c/c

REFERENCE

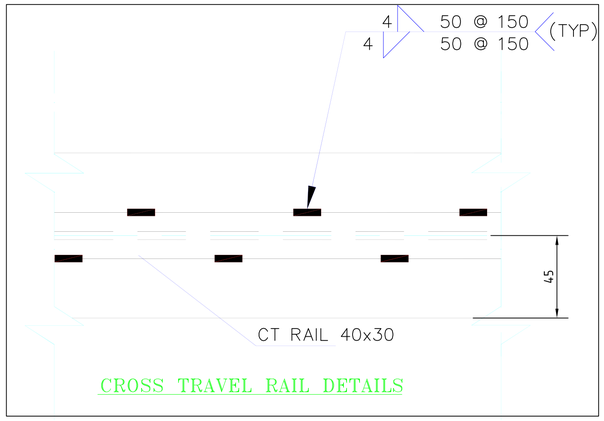
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

ATTACHMENT 1 LOAD DATA DRAWING

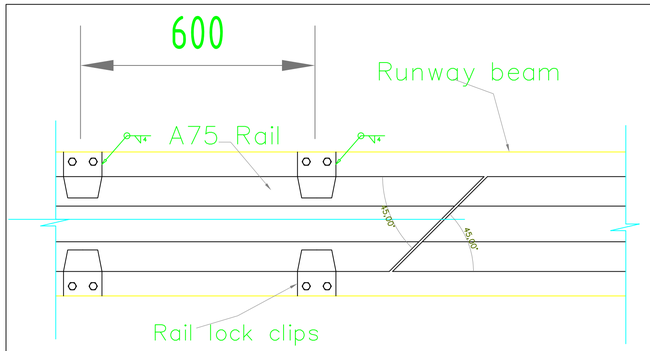
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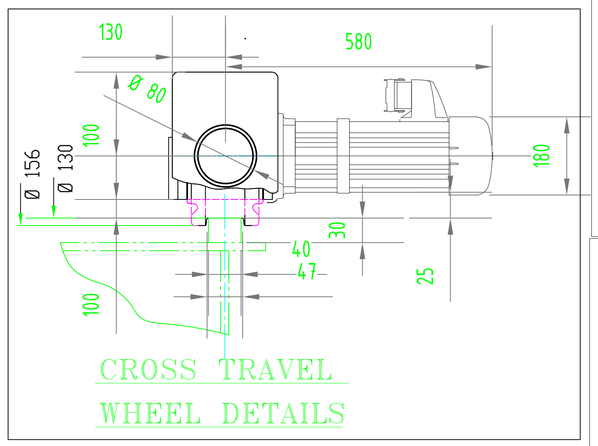
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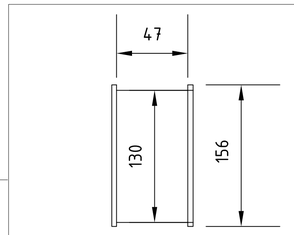
CROSS TRAVEL RAIL DETAILS



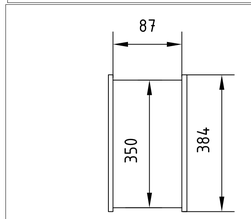
LONG TRAVEL RAIL DETAILS



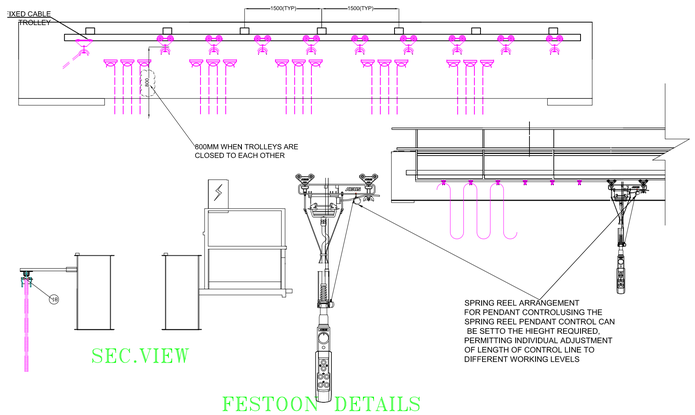
CROSS TRAVEL WHEEL DETAILS



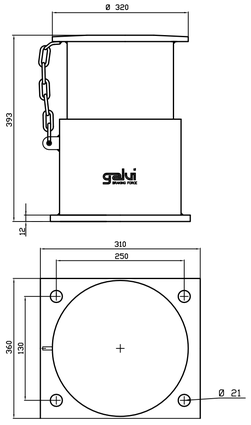
CT WHEEL DETAILS



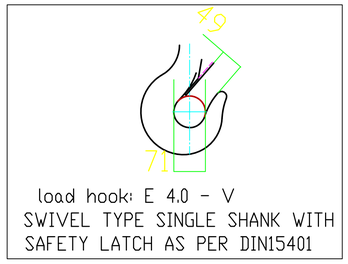
LT WHEEL DETAILS

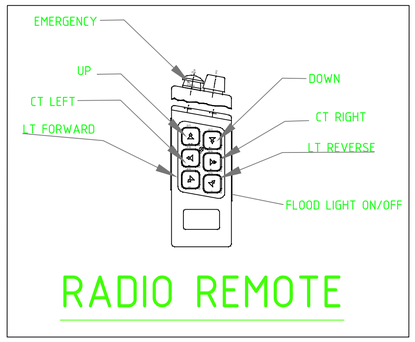


FESTOON DETAILS

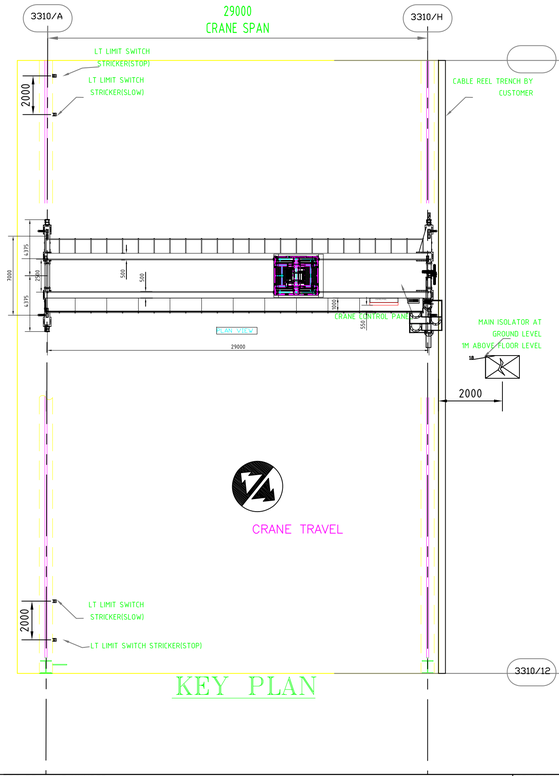


SPRING BUFFER





RADIO REMOTE

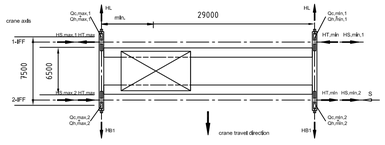


KEY PLAN

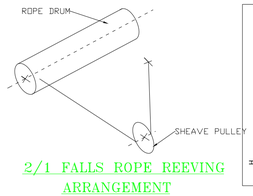
3310-CRN-001
FLOTATION AREA CRANE

Load effects and relevant force components:

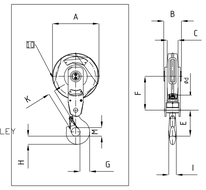
Wheel loads (vertical)	Force component from the mass of the crane and the hoist(s) per crane axis	Qc,min.1	30,7	Qc,max.1	41,6	[kN]
		Qc,min.2	37,6	Qc,max.2	48,6	[kN]
	Force component from mass of hoist load per crane axis	Qh,min.1	5,9	Qh,max.1	29,7	[kN]
		Qh,min.2	5,8	Qh,max.2	29,7	[kN]
Lateral loads (horizontal)	Force from the acceleration of the crane with hoist load (mass force)	HT,min	5,7	HT,max	7,8	[kN]
	skewing force (coefficient of frictional contact = 0,3)	S			53,2	[kN]
	Horizontal force from skewing per crane axis	HS,min.1	0,0	HS,max.1	0,0	[kN]
		HS,min.2	3,3	HS,max.2	45,9	[kN]
Longitudinal loads (horizontal) (per runway beam)	Force from the acceleration of the crane with hoist load (mass force)	HL			11,8	[kN]
	Force from impact on buffer (collision force) (motion limiter taken into account)	HB1			31,1	[kN]



LONG TRAVEL WHEEL LOADS FOR
CRANE CAPACITY - 7.5 T, CRANE SPAN - 29000 MILLIMETER
TYPE OF CRANE - DOUBLE GIRDER GANTRY CRANE
TOTAL NUMBER OF LONG TRAVEL WHEELS - 08 NUMBERS(MAX WHEEL LOAD = 78.5 kN)
SAIZE OF LONG TRAVEL RAILS - A75 AS PER DIN 536

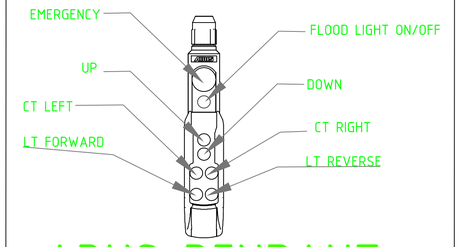


2/1 FALLS ROPE REEVING ARRANGEMENT



HOOK BLOCK TABLE

EQUIPMENT NUMBER	REEVING	DIMENSION "A" (MM)	DIMENSION "B" (MM)	DIMENSION "C" (MM)	DIMENSION "D" (MM)	DIMENSION "E" (MM)	DIMENSION "F" (MM)	DIMENSION "G" (MM)	DIMENSION "H" (MM)	DIMENSION "I" (MM)	DIMENSION "K" (MM)	DIMENSION "M" (MM)	WIRE ROPE DIA ø(MM)
3310-CRN-001	2 / 1	384	138	90	310	250	280	82	67	63	49	71	14



ABUS PENDANT

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3310-EG-ME-GAD-1047	GENERAL ARRANGEMENT ROUGHER AND SCAVENGER/CLEANER FLOTATION SECTION G-G								
MD-216-3310-EG-ME-GAD-1040	GENERAL ARRANGEMENT ROUGHER AND SCAVENGER/CLEANER FLOTATION PLAN VIEW								
MD-216-3300-EG-ME-DAT-V557-1001	DATA SHEET FLOTATION AREA GANTRY CRANE	A03	05/07/2020	ISSUED FOR APPROVAL					
MD-216-3300-EG-ME-GAD-V557-1004	GENERAL ARRANGEMENT FLOTATION AREA CRANE MAJOR COMPONENTS AND ACCESSORIES	A02	16/06/2020	ISSUED FOR APPROVAL					
MD-216-3300-EG-ME-GAD-V557-1002	GENERAL ARRANGEMENT FLOTATION AREA CRANE PLAN AND ELEVATION VIEW	A01	02/04/2020	ISSUED FOR APPROVAL	--	RKM	--	NBR	

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E info@acecranes.com; W www.acecranes.com

DRAWING NUMBER					REVISION
LT102156_SK_GA_CR2.1					02
DATE	03/07/2020	DATE	05/07/2020	DATE	05/072020
DRAWN	SAN	CHECKED	KGN.	APPROVED	GRU

LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_255_J_G_17562_001	2
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707__GAU01__ZF01__MTL01__33006	OU100682142 / 2

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DRAWN	--	--
CHECKED	RKM	02/04/2020
ENGINEER	--	--
APPROVED	NBR	02/04/2020
EMPLOYER		
SCALE	1:125	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
FLOATATION AREA
CRANE PLAN & ELEVATION VIEW

DRAWING NUMBER	REV
MD-216-3300-EG-ME-GAD-V557-1003	A03